Holding the Right Stock: Applying Inventory Management Techniques to the High-Inventory Retail Model of MR D.I.Y. Group (M) Berhad

Nur Fatin Liyana binti Mohd Fazil (E20251036252)

Abstract

MR D.I.Y. Group (M) Berhad is Malaysia's largest home improvement retailer, and its whole business rests on a simple promise: the item you need will be on the shelf, at a low price. Keeping that promise is expensive. Between FY2021 and FY2025 the group grew its network from 900 to 1,556 stores and lifted revenue from RM3.37 billion to about RM4.95 billion. Over the same period, though, inventory turnover days, the number of days of stock the company holds measured against cost of sales, rose from 133 days to a peak of 167 days in FY2023 before settling at 159 days in FY2025. By the end of FY2025, inventories stood at RM1.18 billion, which is roughly five months of goods sitting in warehouses and stores.

This case uses inventory management concepts from Principles of Operations Management to ask whether that level of stock is a sensible buffer or an avoidable cost. Using the company's own disclosures, the analysis shows that average inventory rose 4.4 percent in FY2025 while cost of sales grew only 3.2 percent, so turnover slowed slightly from 2.32 to 2.30 times a year. An illustrative estimate suggests that every 10 days of excess cover ties up around RM71 million in working capital. ABC analysis is then used to show how a single, uniform stock policy across some 17,000 stock-keeping units (SKUs) can leave too much money sitting in slow-moving items.

The case recommends that MR D.I.Y. classify its range using ABC analysis, set safety stock by item class rather than across the board, use its automated warehouse to replenish stores more often in smaller quantities, and track inventory days at store and category level. Taken together, the case shows that deliberate inventory decisions, and not only sales growth, shape a retailer's cash position and its resilience.

Keywords: *operations management; inventory management; ABC analysis; safety stock; days inventory outstanding; inventory turnover; MR D.I.Y.; retail*

1.0 Introduction

Retailers that sell thousands of low-priced items live with an awkward trade-off. If a customer walks in for a particular screwdriver, extension cord or storage box and finds an empty shelf, the sale is usually lost, and that customer may not come back. Yet every extra carton held “just in case” ties up cash, takes up space and runs the risk of being damaged or going out of fashion. For a value retailer that earns a thin margin on each item, getting this balance wrong in either direction soon shows up in the accounts. MR D.I.Y. Group (M) Berhad is a useful example of a company that has deliberately chosen to lean towards holding more stock, and of the costs that come with that choice.

MR D.I.Y. was chosen for this article for three reasons. First, it is a household name in Malaysia with more than 1,500 stores, so its inventory decisions affect suppliers, logistics providers and millions of shoppers, not just a single outlet. Second, because it has been listed on the Main Market of Bursa Malaysia since October 2020, the company publishes quarterly results, annual reports and a five-year financial highlights table that includes its own measure of inventory turnover days. That gives this study a reliable, checkable set of figures without needing access to internal records. Third, the period from FY2021 to FY2025 covers pandemic-era supply disruption, rapid store expansion and the commissioning of a new automated warehouse, which together make it a rich setting in which to apply and test inventory management theory.

The company began in July 2005 as a single hardware shop on Jalan Tuanku Abdul Rahman in Kuala Lumpur, founded by Tan Yu Yeh (Wikipedia, 2026). It has since grown into a multi-brand value retail group that runs MR D.I.Y., MR. TOY and MR. DOLLAR stores, among other formats, and carries a range of roughly 17,000 SKUs covering hardware, household goods, electrical items, stationery, toys and personal care (MR D.I.Y. Group, 2024; MBSB Research, 2025). Its 2020 listing raised about RM1.5 billion, making it Malaysia's largest initial public offering that year (MR D.I.Y., 2020). Before the listing, revenue had already grown from RM1.23 billion in FY2017 to RM2.28 billion in FY2019 (Value Invest Asia, n.d.). As at 31 December 2025, the group operated 1,556 stores and reported revenue of about RM4.95 billion and profit after tax of RM632.69 million (Bernama, 2026).

The aim of this article is to use inventory management techniques to explain why MR D.I.Y. carries such a large stock position, to measure how efficiently that stock is being used with the

company's own reported figures, and to suggest practical ways of freeing up working capital without hurting product availability. Section 2.0 sets out the managerial problem, Section 3.0 applies the chosen technique, Section 4.0 considers the implications for practitioners and academics, and Section 5.0 concludes.

2.0 Problem Statement

The managerial problem examined here is that MR D.I.Y. holds a very large, slowly turning inventory relative to its sales. This ties up well over a billion ringgit of working capital and exposes the company to holding, handling and obsolescence costs. As the researcher did not have access to internal planning data, the problem was defined from the company's published annual reports, quarterly financial statements and press releases, supported by business press coverage and equity research reports rather than interviews.

Three pieces of evidence show why this matters. First, inventory turnover days have drifted upwards. According to the company's financial highlights, stock cover rose from 133 days in FY2021 to 144 days in FY2022 and 167 days in FY2023, before easing to 157 days in FY2024 and edging back up to 159 days in FY2025 (MR D.I.Y. Group, 2026a). Put simply, an item bought today will wait, on average, more than five months before it is sold. Second, the amount of money tied up is large and still growing. Inventories stood at RM1,084.40 million at the end of FY2023 and RM1,082.45 million at the end of FY2024, then jumped to RM1,179.03 million at the end of FY2025, an increase of 8.9 percent in a year when cost of sales grew by only 3.2 percent (MR D.I.Y. Group, 2025b, 2026b). Third, the physical flow of goods has not always kept pace. Analysts linked weaker same-store sales in the fourth quarter of FY2024 partly to integration problems at the group's new automated warehouse and its inventory system, whose full start-up had slipped from the original November 2024 target (Kenanga Research, 2025). In other words, holding plenty of stock did not, on its own, guarantee that the right goods reached the right stores.

Part of the explanation is a conscious management choice. In its 2023 annual report, the company explained that inventory turnover days had been kept at slightly higher levels by design, after supply shortages during the pandemic showed how damaging empty shelves could be (MR D.I.Y. Group, 2024). That is a reasonable response to uncertain lead times. Much of the merchandise is bought in bulk and moved through long supply routes, so a single delayed shipment can leave

hundreds of stores short at once, and the pandemic showed how quickly global supply chains can seize up (Ivanov, 2020). The difficulty is that a buffer which made sense when freight was unreliable can easily become a permanent habit once conditions return to normal. Research on retail operations has long shown that inventory turnover varies widely between retailers and is shaped by strategic choices as well as by sales (Gaur et al., 2005), and that extra inventory lifts sales only up to a point, after which it adds cost and clutter without adding much revenue (Ton & Raman, 2010).

Left unchecked, the problem carries real costs. Money held in stock cannot be used to open stores, pay dividends or reduce debt, and goods that sit too long risk damage, obsolescence and markdowns, particularly in seasonal and trend-driven lines such as toys. A large stock position also puts pressure on warehouse and shelf space and makes it harder to see which items are actually selling. As Section 3.4 shows, even a modest cut in inventory days would release tens of millions of ringgit, a meaningful sum for a business whose annual profit is around RM630 million.

3.0 Applications of Operations Management Technique

3.1 Choice of Technique: Inventory Management

Inventory management is the operations management technique concerned with how much raw material, work-in-process and finished goods a business should hold, and when it should replenish them, so that it avoids stock-outs without carrying unnecessary holding cost (Heizer et al., 2023). For a retailer, almost all inventory is finished goods bought for resale, so the technique applies directly to the problem described in Section 2.0. It was judged more relevant here than other techniques covered in the course. Forecasting would help predict demand for individual products, and layout strategy could improve how stores and warehouses are organised, but the core issue at MR D.I.Y., namely how much stock to hold and of which items, is first and foremost an inventory decision.

Three inventory concepts are applied in this article. The first is ABC analysis, which ranks items by their annual monetary volume so that managers can focus tight control on the small number of items that account for most of the value (Heizer et al., 2023; Ramanathan, 2006). The second is safety stock and service-level thinking, which explains why a buffer is needed when demand and lead times are uncertain, and why that buffer should differ from one item to another. The third is a

pair of standard performance ratios, days inventory outstanding (DIO) and inventory turnover, calculated from MR D.I.Y.'s reported inventory and cost of sales to measure how efficiently stock is turned into sales. These tools are widely used in retail operations research and practice (Gaur et al., 2005; Silver et al., 2017).

3.2 Data and Assumptions

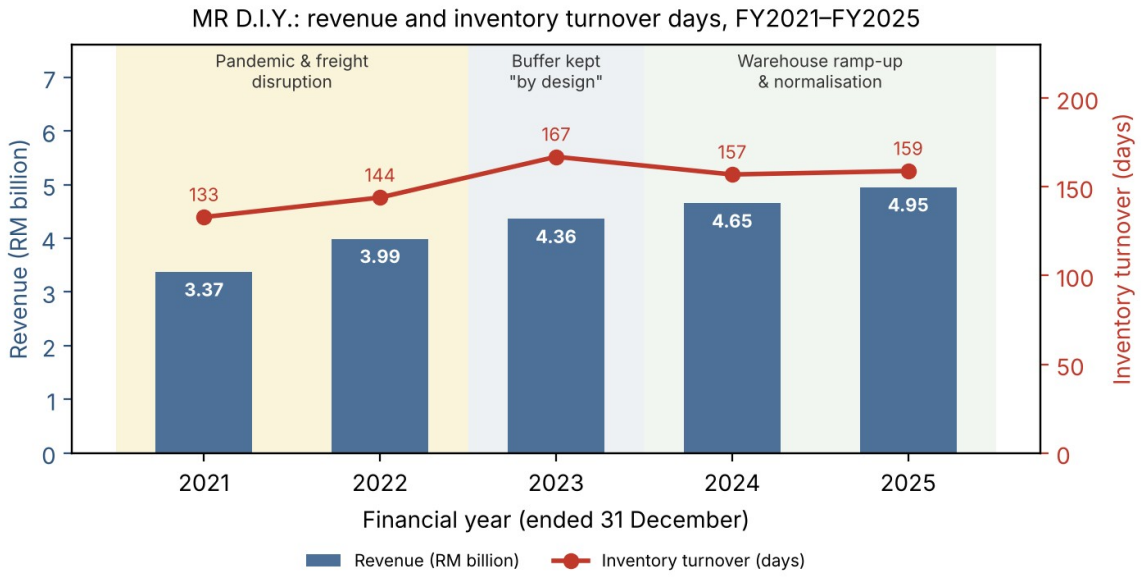
Table 1 and Figure 1 summarise MR D.I.Y.'s revenue, profit after tax, store count and inventory turnover days for FY2021 to FY2025 (financial year ended 31 December). The figures are taken from the company's results announcements and financial highlights, together with business press reports of those results.

Table 1. MR D.I.Y. revenue, profit after tax, store network and inventory turnover days, FY2021–FY2025

Financial year	Revenue (RM billion)	Profit after tax	Stores at year-end	Inventory turnover (days)	Main event
FY2021	3.37	RM431.83 million	900	133	Pandemic movement restrictions; continued store roll-out
FY2022	3.99	RM472.95 million	1,080	144	Global freight and supply-chain strain; stock built up as a buffer
FY2023	4.36	RM560.68 million (record)	1,261	167	Inventory days kept higher “by design”
FY2024	4.65	RM568.94 million	1,435	157	Automated warehouse integration issues in 4Q; weaker same-store sales
FY2025	4.95	RM632.69 million (record)	1,556	159	121 net new stores; inventories rise to RM1.18 billion

Sources: MR D.I.Y. Group (2023, 2024, 2025a, 2026a); Malay Mail (2023, 2024); New Straits Times (2025); Bernama (2026); Business Today (2026); Kenanga Research (2025).

Figure 1. MR D.I.Y. revenue and inventory turnover days, FY2021–FY2025



Note. Prepared by the author from the data in Table 1.

3.3 ABC Analysis: Not Every SKU Deserves the Same Stock Policy

ABC analysis applies the Pareto principle to inventory. Items are ranked by annual monetary volume, that is, annual demand multiplied by unit cost, and then sorted into three groups. Class A items make up a small share of the range but account for most of the value; Class B items sit in the middle; and Class C items form the bulk of the range but only a small share of the value (Heizer et al., 2023). The point of the exercise is to spend management attention where the money is. Class A items deserve frequent review, accurate forecasts and tight stock control, whereas Class C items can be handled with simpler rules and larger, less frequent orders.

MR D.I.Y.'s SKU-level sales data are not public, so Table 2 applies typical textbook proportions to the company's range of about 17,000 SKUs and its FY2025 cost of sales of RM2,599.75 million. The figures are illustrative rather than actual, but they give a sense of the scale involved.

Table 2. Illustrative ABC classification of MR D.I.Y.'s product range

Class	Share of SKUs (approx. number)	Share of annual cost-of-sales value (approx. RM million)	Suggested inventory policy
A	About 15% (≈ 2,550 SKUs)	About 75% (≈ RM1,950 million)	Frequent or continuous review; tight safety stock; frequent replenishment from the central warehouse
B	About 30% (≈ 5,100 SKUs)	About 20% (≈ RM520 million)	Periodic review; moderate safety stock
C	About 55% (≈ 9,350 SKUs)	About 5% (≈ RM130 million)	Simple reorder rules; larger, less frequent orders; avoid overbuying slow and seasonal lines

Note. Proportions follow the typical ABC split described in Heizer et al. (2023) and are applied by the author for illustration only. The 17,000-SKU range is from MR D.I.Y. Group (2024) and cost of sales from MR D.I.Y. Group (2026b).

The lesson for MR D.I.Y. is that a company-wide figure such as “about 160 days of cover” hides very different situations. If most of the cash sits in a few thousand fast-moving A items, then cutting their cover by even a few weeks, backed by more frequent deliveries, would release far more money than trimming thousands of low-value C items. On the other hand, generous cover on cheap C items costs relatively little and protects the store's reputation for always having the small things customers come in for. Safety stock follows the same logic. The standard formula, safety stock = $Z \times \sigma dLT$, shows that the buffer depends on the service level chosen (Z) and on how much demand varies during the replenishment lead time (σdLT) (Heizer et al., 2023). A faster, more reliable warehouse shortens lead times and reduces that variation, which means the same service level can be reached with less stock.

3.4 Inventory Performance Ratios: Days Inventory Outstanding and Turnover

To measure the problem directly, this article calculates DIO and inventory turnover for FY2024 and FY2025 from the company's quarterly financial statements. Following the company's own definition, DIO is average inventory divided by cost of sales, multiplied by the number of days in the year, and inventory turnover is cost of sales divided by average inventory (MR D.I.Y. Group, 2026a). Because it uses cost of sales rather than revenue, this method gives a truer picture of how long goods actually sit before they are sold. The calculated values also match the company's reported figures of 157 and 159 days, which gives some confidence in the data.

Table 3. MR D.I.Y. inventory position, FY2024 and FY2025

Measure	FY2024 (as at 31 Dec 2024)	FY2025 (as at 31 Dec 2025)	Change
Closing inventories (RM million)	1,082.45	1,179.03	+8.9%
Average inventories (RM million)	1,083.42	1,130.74	+4.4%
Cost of sales (RM million)	2,518.77	2,599.75	+3.2%
Stores at year-end	1,435	1,556	+8.4%
Closing inventory per store (RM million)	0.754	0.758	+0.5%
Days inventory outstanding	157.4 days	158.8 days	+1.4 days
Inventory turnover	2.32 times/year	2.30 times/year	−0.02 times/year

Note. Inventory balances and cost of sales from MR D.I.Y.'s unaudited fourth-quarter reports for FY2024 and FY2025 (MR D.I.Y. Group, 2025b, 2026b); average inventory for FY2024 uses the opening balance of RM1,084.40 million at 31 December 2023. DIO calculated by the author as (Average inventory ÷ Cost of sales) × 366 for FY2024 and × 365 for FY2025; turnover as Cost of sales ÷ Average inventory.

At first glance, Table 3 points to a stable position. DIO moved only from about 157 to 159 days, and inventory per store barely changed, which suggests that much of the extra stock simply followed the 121 new stores opened during the year. Look a little closer, though, and the picture is less comfortable. Closing inventory grew almost three times as fast as cost of sales (8.9 percent against 3.2 percent). Measured on the year-end balance alone, the group went into FY2026 holding about 165.5 days of cover, noticeably more than the 159-day annual average. Stock is building faster than sales, and that is exactly the kind of early warning inventory management theory tells managers to watch for (Silver et al., 2017).

Holding cost puts these numbers in perspective. Holding cost rates differ from firm to firm, so this article assumes 20 percent a year purely for illustration, covering the cost of capital, storage, handling, shrinkage and obsolescence. On that basis, carrying an average inventory of RM1,130.74 million costs MR D.I.Y. roughly RM226 million a year, equivalent to more than a third of its FY2025 profit after tax. Each 10-day reduction in DIO at FY2025 cost of sales would release about RM71.2 million of working capital and save about RM14.2 million a year in holding cost. A return to the FY2021 level of 133 days, a fall of about 26 days, would free up roughly RM185 million. These are rough estimates, but they show that the company has a strong financial reason to manage stock more tightly now that the supply shocks of 2021 and 2022 have eased.

3.5 Managerial Recommendations Derived from the Analysis

- **Classify the range with ABC analysis.** MR D.I.Y. should rank its roughly 17,000 SKUs by annual monetary volume and refresh the ranking every quarter, so that stock targets, review frequency and buyers' attention are concentrated on the items that hold most of the money.
- **Set safety stock by item class, not across the board.** Instead of one company-wide buffer carried over from the pandemic years, safety stock should be set by class, with higher service levels for essential, fast-moving A items and leaner cover for low-value or seasonal C items, where a brief stock-out does little harm.
- **Use the automated warehouse to replenish little and often.** Once the new automated warehouse is running at full capacity, smaller and more frequent deliveries to stores would reduce the stock each outlet needs to hold, shorten replenishment lead times and dampen the demand variation that drives safety stock.
- **Review the pandemic-era buffer against current lead times.** Because the higher inventory days were kept by design when freight was unreliable, the policy should be revisited using up-to-date supplier lead-time data and scaled back step by step wherever conditions have normalised.
- **Track inventory days by store and category, not only at group level.** As Table 3 shows, a steady group DIO can hide stock that is building faster than sales. Monthly DIO and turnover by category and by store, read alongside the absolute inventory figure, would help managers spot slow-moving lines before they need to be marked down.

4.0 Implications of Operations Management

4.1 Implications for Practitioners

For MR D.I.Y.'s operations and merchandising managers, and for managers at other multi-store value retailers, this analysis shows that the amount of stock a company holds is a management decision with a measurable price, not just a by-product of growth. The DIO and turnover ratios in Table 3 can be produced every quarter at almost no cost, and they reveal trends, such as inventory growing faster than cost of sales, that the headline inventory figure does not. The ABC illustration

in Table 2 offers a practical place to start. By separating the few items that carry most of the value from the many that carry very little, managers can cut working capital where it matters most while keeping the small, everyday items that bring customers through the door reliably on the shelf. More broadly, the case shows that investments in distribution, such as an automated warehouse, only deliver their full value when stocking policies are revised to take advantage of shorter and more reliable lead times.

4.2 Implications for Theorists and Academics

For operations management scholars, MR D.I.Y. offers a large, well-documented example of a retailer that consciously traded higher inventory for supply security after the COVID-19 pandemic. This supports empirical work showing that inventory turnover in retail reflects strategic choices as well as sales performance (Gaur et al., 2005), and that the link between inventory and sales has diminishing returns (Ton & Raman, 2010). The case also points to a gap in the literature. A great deal has been written about how the pandemic disrupted global supply chains (Ivanov, 2020), but far less about how and when firms should unwind the buffers they built in response once conditions settle. Finally, the case gives lecturers a concrete, numerical illustration of ABC analysis and safety stock applied to a familiar Malaysian retailer, which may help students connect textbook formulas with real financial results.

4.3 Lessons Learnt

Three lessons stand out from this study. First, a safety buffer built in response to a crisis should come with a review date. MR D.I.Y.'s decision to hold more stock during the pandemic was sensible, but without a deliberate plan to unwind it, the buffer risks hardening into a permanent cost. Second, averages can mislead. A group-wide inventory figure, or even a steady DIO, can hide stock piling up in particular categories or stores, which is why ABC analysis and more detailed ratios matter. Third, physical flow and inventory policy go hand in hand. The warehouse difficulties of late FY2024 showed that having plenty of stock is not the same as having the right stock in the right place; what really counts is getting the right items to the right stores at the right time.

5.0 Conclusion

This article examined MR D.I.Y. Group (M) Berhad, Malaysia's largest home improvement retailer, and identified its high and slowly turning inventory as a critical managerial problem, based on the company's own published financial disclosures. Applying inventory management as the relevant operations management technique, the analysis showed that inventory turnover days rose from 133 in FY2021 to a peak of 167 in FY2023 and still stood at 159 in FY2025, while year-end inventories climbed to RM1,179.03 million. Calculations from the company's reported figures confirmed that average inventory grew faster than cost of sales in FY2025, and an illustrative holding-cost estimate suggested that each 10-day cut in inventory days would release around RM71 million of working capital. An illustrative ABC classification of the company's roughly 17,000 SKUs showed why a single, uniform stock policy is unlikely to be efficient. These findings support five practical recommendations: classify the range using ABC analysis, set safety stock by item class, use the automated warehouse to replenish stores little and often, review the pandemic-era buffer against current lead times, and track inventory days by store and category.

The main limitation of this study is its reliance on publicly available financial statements rather than internal operational data. The ABC classification in Section 3.3 uses typical textbook proportions instead of actual SKU-level sales, and the 20 percent holding cost rate is an assumption, since MR D.I.Y.'s real carrying costs are not disclosed. Future research could use store- or category-level data, if the company were willing to share it, to test these estimates, and could compare MR D.I.Y. with other Malaysian and regional retailers to see whether its inventory position is unusual for the sector. Even so, the analysis shows that classical inventory management tools can shed real light on the operational and financial choices of one of Malaysia's best-known retailers.

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Appendix

Appendix A: Inventory Management Formulas Used

ABC classification: Annual monetary volume = Annual demand (units) × Unit cost (RM); items are ranked from highest to lowest and grouped into Class A (about 70–80% of value), Class B (about 15–25%) and Class C (about 5%)

Safety stock: SS = Z × σdLT, where Z = number of standard deviations for the chosen service level and σdLT = standard deviation of demand during the replenishment lead time

Days inventory outstanding (company definition): DIO = (Average inventory ÷ Cost of sales) × number of days in the year, where Average inventory = (Opening inventory + Closing inventory) ÷ 2

Inventory turnover: Turnover = Cost of sales ÷ Average inventory

Annual holding cost (illustrative): Holding cost = Average inventory × Holding cost rate (assumed at 20% per year)

Appendix B: Worked Calculation of Days Inventory Outstanding, Turnover and Holding Cost

FY2024: Average inventory = (RM1,084.40 million + RM1,082.45 million) ÷ 2 = RM1,083.42 million. DIO = (RM1,083.42 million ÷ RM2,518.77 million) × 366 ≈ 157.4 days. Turnover = RM2,518.77 million ÷ RM1,083.42 million ≈ 2.32 times per year.

FY2025: Average inventory = (RM1,082.45 million + RM1,179.03 million) ÷ 2 = RM1,130.74 million. DIO = (RM1,130.74 million ÷ RM2,599.75 million) × 365 ≈ 158.8 days. Turnover = RM2,599.75 million ÷ RM1,130.74 million ≈ 2.30 times per year.

Holding cost and working capital (illustrative): Annual holding cost ≈ RM1,130.74 million × 20% ≈ RM226.1 million. Inventory released per 10-day cut in DIO ≈ (RM2,599.75 million ÷ 365) × 10 ≈ RM71.2 million, saving about RM14.2 million a year in holding cost. A 26-day cut (159 to 133 days) ≈ RM185.2 million released.

Change: closing inventory rose by 8.9% while cost of sales rose by only 3.2%. On the closing balance alone, FY2025 cover = (RM1,179.03 million ÷ RM2,599.75 million) × 365 ≈ 165.5 days, above the average-based 158.8 days, confirming that stock built up faster than sales during the year.